ECE316 Formula

Continuous-Time Fourier Series (CTFS) Properties

For a periodic signal $x(t)$ with fundamental period T_0, fundamental frequency $\omega_0 = \frac{2\pi}{T_0}$, and Fourier Series coefficients $\alpha_k = \frac{1}{T_0}\int_{T_0} x(t)e^{-j\omega_0 kt}dt$, the properties are:

Operation	Time Domain ($x(t)$)	Frequency Domain (α_k)
Time-shift by t_0	$x(t-t_0)$	$e^{-j\omega_0 k t_0}\alpha_k$
Frequency-shift by k_0	$e^{j\omega_0 k_0 t}x(t)$	α_{k-k_0}
Conjugation	$x(t)^*$	α_{-k}^*
Time-reversal	$x(-t)$	α_{-k}
Differentiation	$\dot{x}(t)$	$(j\omega_0 k)\alpha_k$
Convolution	$\frac{1}{T_0}\int_{T_0} x(\tau)y(t-\tau)d\tau$	$\alpha_k\beta_k$
Multiplication	$x(t)y(t)$	$\sum_{\ell=-\infty}^{\infty}\alpha_\ell\beta_{k-\ell}$
Real Part	$\mathrm{Re}\{x(t)\}$	$\frac{1}{2}(\alpha_k + \alpha_{-k}^*)$
Imag Part	$\mathrm{Im}\{x(t)\}$	$\frac{1}{2j}(\alpha_k - \alpha_{-k}^*)$

Continuous-Time Fourier Transform (CTFT) Properties and Pairs

For a signal $x(t)$ with spectrum $X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$, the properties are:

Properties

Operation	Time Domain ($x(t)$)	Frequency Domain ($X(j\omega)$)
Time-shift by t_0	$x(t-t_0)$	$e^{-j\omega t_0}X(j\omega)$
Frequency-shift by ω_0	$e^{j\omega_0 t}x(t)$	$X(j(\omega-\omega_0))$
Conjugation	$x(t)^*$	$X(-j\omega)^*$
Time-scaling	$x(at)$	$\frac{1}{\lvert a\rvert}X\left(j\frac{\omega}{a}\right)$
Differentiation	$\dot{x}(t)$	$(j\omega)X(j\omega)$
Convolution	$\int_{-\infty}^{\infty} x(\tau)y(t-\tau)d\tau$	$X(j\omega)Y(j\omega)$
Multiplication	$x(t)y(t)$	$\frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\nu)Y(j(\omega-\nu))d\nu$
Real Part	$\mathrm{Re}\{x(t)\}$	$\frac{1}{2}(X(j\omega) + X(-j\omega)^*)$
Imag Part	$\mathrm{Im}\{x(t)\}$	$\frac{1}{2j}(X(j\omega) - X(-j\omega)^*)$

Useful Pairs

Time Domain ($x(t)$)	Frequency Domain ($X(j\omega)$)
$\delta(t)$	1
$e^{j\omega_0 t}$	$2\pi\delta(\omega-\omega_0)$
$e^{-at}u(t)$ with $a>0$	$\frac{1}{j\omega+a}$
$\frac{\sin(at)}{\pi t}$ with $a>0$	$u(\omega+a)-u(\omega-a)$

Euler Identities

$$e^{j\theta} = \cos\theta + j\sin\theta, \quad \cos\theta = \frac{e^{j\theta}+e^{-j\theta}}{2}, \quad \sin\theta = \frac{e^{j\theta}-e^{-j\theta}}{2j}$$

Half-Wave Rectified Circuit Output

$$y(t) = \begin{cases} \cos(120\pi t) & \text{if } \cos(120\pi t) \geq 0 \\ 0 & \text{if } \cos(120\pi t) < 0 \end{cases}$$

Basically just remove everything under the x-axis.

Average Power of a signal $x(t)$ is defined as

$$P_x = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} \left|x^2(t)\right| dt$$

The energy of $x(t)$ is

$$E_x = \lim_{T\to\infty} \int_{-T}^{T} \left|x^2(t)\right| dt = \int_{-\infty}^{\infty} \left|x^2(t)\right| dt$$

- $x(t)$ is an energy signal iff $0 \le E < \infty$; i.e. $x(t)$ has finite energy. （iff means if and only if）
- $x(t)$ is an power signal iff $0 < P < \infty$; i.e. $x(t)$ has finite power. Note that $E = \infty$ for power signals.

Sinusoidal Signal

$$x(t) = Acos(2\pi f_c t + \theta)$$

where A is amplitude, f_c is frequency (in Hz), and θ is phase

Rectangular Function

$$A \cdot \text{rect}(\frac{t - t_0}{T})$$

where A is the hight, t_0 is the centre, T is the width.

Fourier Series

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{j\omega_0 kt}, \quad c_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-j\omega_0 kt} dt, \quad \frac{1}{T_0} \|x\|^2 = \sum_{k=-\infty}^{\infty} |c_k|^2$$

Angle Sum and Difference Identity

$$\cos(A+B) = \cos(A)\cos(B) \mp \sin(A)\sin(B)$$
$$\sin(A+B) = \sin(A)\cos(B) \pm \cos(A)\sin(B)$$

Sinc Function

$$\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$$

> *Note that* sinc *function have different definitions, in the course we use this definition.*

Parseval's Theorem

$$P_x = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |X_K|^2$$

Rayleigh Energy Theorem

Rayleigh Energy Theorem is a continuous version of Parseval Theorem.

The **energy** of a signal $g(t)$ is given by:

$$E = \int_{-\infty}^{\infty} |g(t)|^2 \, dt = \int_{-\infty}^{\infty} |G(f)|^2 \, df$$

(in Joules/Hz)

Let:

$$\psi_g(f) = |G(f)|^2$$

Then $\psi_g(f)$ is called the **energy spectral density**, because the energy of $g(t)$ equals:

$$E = \int_{-\infty}^{\infty} \psi_g(f) \, df$$

Fourier Transform

$$G(f) = \int_{-\infty}^{\infty} g(\lambda) e^{-j2\pi f_0 \lambda} d\lambda$$

Inverse Fourier Transform

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{j2\pi f_0 t} df$$

where $\omega = 2\pi f_0$, so can replace $2\pi f_0$ with ω. **BUT IN THIS FUCKING COURSE, THE TA SAID WE SHOULD USE $2\pi f_0$...**

Properties of the Fourier Transform

To simplify the calculation of Fourier transforms, we often use shortcuts based on the following properties:

1. Superposition

$$x_1(t) + x_2(t) \leftrightarrow X_1(f) + X_2(f)$$

2. Time Delay

$$x(t - t_0) \leftrightarrow e^{-j2\pi f_0 t} X(f)$$

3. Scale Change

$$x(at) \leftrightarrow \frac{1}{|a|} X\left(\frac{f}{a}\right)$$

4. Duality

$$X(t) \leftrightarrow x(-f)$$

> *Think of a function in terms of its graph.*

5. Frequency Translation

$$x(t) e^{j2\pi f_0 t} \leftrightarrow X(f - f_0)$$

6. Modulation

$$x(t) \cos(2\pi f_0 t) \leftrightarrow \frac{1}{2}[X(f - f_0) + X(f + f_0)]$$

7. Differentiation

$$\frac{d}{dt} x(t) \leftrightarrow j2\pi f X(f)$$

8. Convolution

Time domain:

$$x_1(t) * x_2(t) = \int_{-\infty}^{\infty} x_1(\tau) x_2(t - \tau)\, d\tau$$

Frequency domain:

$$x_1(t) * x_2(t) \leftrightarrow X_1(f) X_2(f)$$

9. Multiplication

$$x_1(t) x_2(t) \leftrightarrow X_1(f) * X_2(f)$$

10. Delta Function

$$\delta(t) \leftrightarrow 1$$

$$e^{j2\pi f_0 t} \leftrightarrow \delta(f - f_0)$$

12. Area Property

> *The area under a function in one domain equals the value of the function at zero in the other domain.*

13. Sinc Function

$$\text{rect}(\frac{t}{T}) \leftrightarrow T\,\text{sinc}(Tf)$$

Signal Bandwith

- **Strict bandwidth**: for signals whose spectrum restricted to some band, e.g. the sinc pulse, or a sum of sinusoids.
- **- 3 dB bandwidth**: The range of frequencies over which the energy spectral density (or power spectral density for power signals) is within 3 dB of the maximum value. This is used mostly in electronic circuit design, e.g. amplifier design.
- **x% bandwidth (e.g. x=99)**. The minimum range of frequencies over which the area under the energy spectral density curve (or power spectral density curve) equals x % of the total area.
- **rms bandwidth**

$$B_{rms} = (\frac{\int_{-\infty}^{\infty} f^2 |X(f)|^2 df}{\int_{-\infty}^{\infty} |X(f)|^2 df})^{\frac{1}{2}}\ \text{Hz}$$

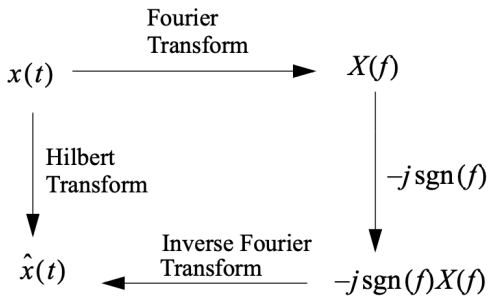
Time Duration of a Signal

$$T_{rms} = (\frac{\int_{-\infty}^{\infty} t^2 |x(t)|^2 dt}{\int_{-\infty}^{\infty} |x(t)|^2 dt})^{\frac{1}{2}}$$

For any signal with rms time duratin T_{rms} and rms bandwidth B_{rms} we have

$$T_{rms} B_{rms} \geq \frac{1}{4\pi}$$

Hilbert Transform



Hilbert transform of $x(t)$ is

$$\hat{x}(t) = \sin(2\pi f_0 t)$$

Signum function

$$\text{sgn}(x) = \frac{x}{|x|}, \quad \text{for } x \neq 0$$

or

$$\text{sgn}(x) = \begin{cases} -1, & \text{if } x < 0 \\ 0, & \text{if } x = 0 \\ 1, & \text{if } x > 0 \end{cases}$$

$$y(t) = h(t) * x(t)$$
$$Y(f) = H(f)X(f)$$

Amplitude Response: $A(f) = |H(f)|$

Phase Response: $\theta(f) = \arg(H(f))$

Energy Spectral Density of a Signal

Energy Spectra:

$$S_x(f) = |X(f)|^2$$

Distortionless Transmission

Non-Linear Distortion

Filters (LP, HP, BP)

To distinguish whether a filter is a high-pass, low-pass, or band-pass filter from its transfer function, you need to analyze the behavior of the transfer function's magnitude, $|H(j\omega)|$, at different frequencies, particularly at the extremes (frequency approaching zero and infinity) and at any potential center frequencies. The variable 's' in the transfer function is the complex frequency, which can be replaced by $j\omega$ for frequency response analysis, where ω is the angular frequency.

Here's how to identify each type of filter:

Low-Pass Filter

A low-pass filter allows low-frequency signals to pass through while attenuating high-frequency signals.

- **Behavior at Low Frequencies** (as $\omega \to 0$): The magnitude $|H(j\omega)|$ will approach a constant, non-zero value. This means the filter passes DC and very low frequencies.
- **Behavior at High Frequencies** (as $\omega \to \infty$): The magnitude $|H(j\omega)|$ will approach zero. This indicates that high frequencies are blocked.

Example Transfer Function:
A simple first-order low-pass filter has a transfer function of the form:
$H(s) = \frac{K}{s+a}$
When you substitute $s = j\omega$:
$H(j\omega) = \frac{K}{j\omega+a}$

- As $\omega \to 0$, $|H(j\omega)| \to \frac{K}{a}$ (a constant).
- As $\omega \to \infty$, $|H(j\omega)| \to 0$.

High-Pass Filter

A high-pass filter allows high-frequency signals to pass through while attenuating low-frequency signals.

- **Behavior at Low Frequencies** (as $\omega \to 0$): The magnitude $|H(j\omega)|$ will approach zero. This means the filter blocks DC and very low frequencies.
- **Behavior at High Frequencies** (as $\omega \to \infty$): The magnitude $|H(j\omega)|$ will approach a constant, non-zero value. This indicates that high frequencies are passed.

Example Transfer Function:
A simple first-order high-pass filter has a transfer function of the form:
$H(s) = \frac{Ks}{s+a}$
When you substitute $s = j\omega$:
$H(j\omega) = \frac{K(j\omega)}{j\omega+a}$

- As $\omega \to 0$, $|H(j\omega)| \to 0$.
- As $\omega \to \infty$, $|H(j\omega)| \to K$ (a constant).

A band-pass filter allows signals within a specific range of frequencies to pass through while attenuating frequencies outside of this range.

- **Behavior at Low Frequencies** (as $\omega \to 0$): The magnitude $|H(j\omega)|$ will approach zero.
- **Behavior at High Frequencies** (as $\omega \to \infty$): The magnitude $|H(j\omega)|$ will also approach zero.
- **Behavior at a Center Frequency** (ω_c): The magnitude $|H(j\omega)|$ will be at its maximum at a specific center or resonant frequency.

Example Transfer Function:
A simple second-order band-pass filter has a transfer function of the form:
$H(s) = \frac{Ks}{s^2+as+b}$
When you substitute $s = j\omega$:
$H(j\omega) = \frac{K(j\omega)}{-\omega^2+aj\omega+b}$

- As $\omega \to 0$, $|H(j\omega)| \to 0$.
- As $\omega \to \infty$, $|H(j\omega)| \to 0$.
- The magnitude will be maximum at a resonant frequency, which is typically related to the coefficients 'a' and 'b'.

AM Modulation

Switching Modulators